In [2]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

In [4]:
```
df = pd.read_csv('data_airquality.csv')
```

```
C:\Users\Dell\AppData\Local\Temp\ipykernel_5328\3334187165.py:1: DtypeWarning: Colum
ns (0: stn_code) have mixed types. Specify dtype option on import or set low_memory=
False.
  df = pd.read_csv('data_airquality.csv')
```

In [12]:
```
df.head()
```

Out[12]:

	stn_code	sampling_date	state	location	agency	type	so2	no2
0	150.0	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	4.8	17.4
1	151.0	February - M021990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	3.1	7.0
2	152.0	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.2	28.5
3	150.0	March - M031990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.3	14.7
4	151.0	March - M031990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	4.7	7.5

In [17]:
```
print(df.columns)
```

```
Index(['stn_code', 'sampling_date', 'state', 'location', 'agency', 'type',
       'so2', 'no2', 'rspm', 'spm', 'location_monitoring_station', 'pm2_5',
       'date'],
      dtype='str')
```

In [29]:
```
df.dtypes
```

Out[29]:
```
stn_code                       object
sampling_date                     str
state                             str
location                          str
agency                            str
type                              str
so2                           float64
no2                           float64
rspm                          float64
spm                           float64
location_monitoring_station       str
pm2_5                         float64
```

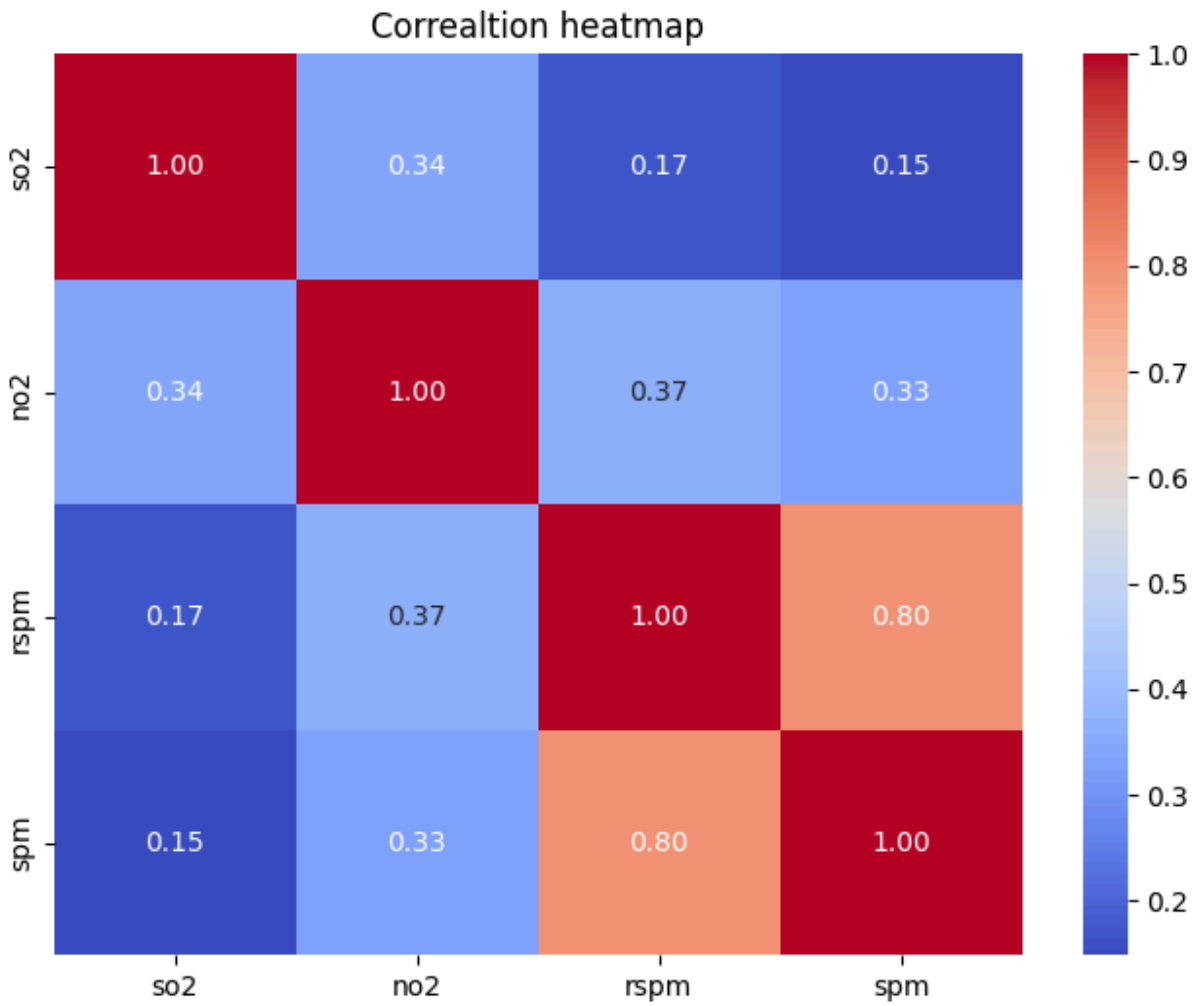
```
date                                 str
dtype: object
```

In [34]:
```
df = df.apply(pd.to_numeric, errors='coerce')
```

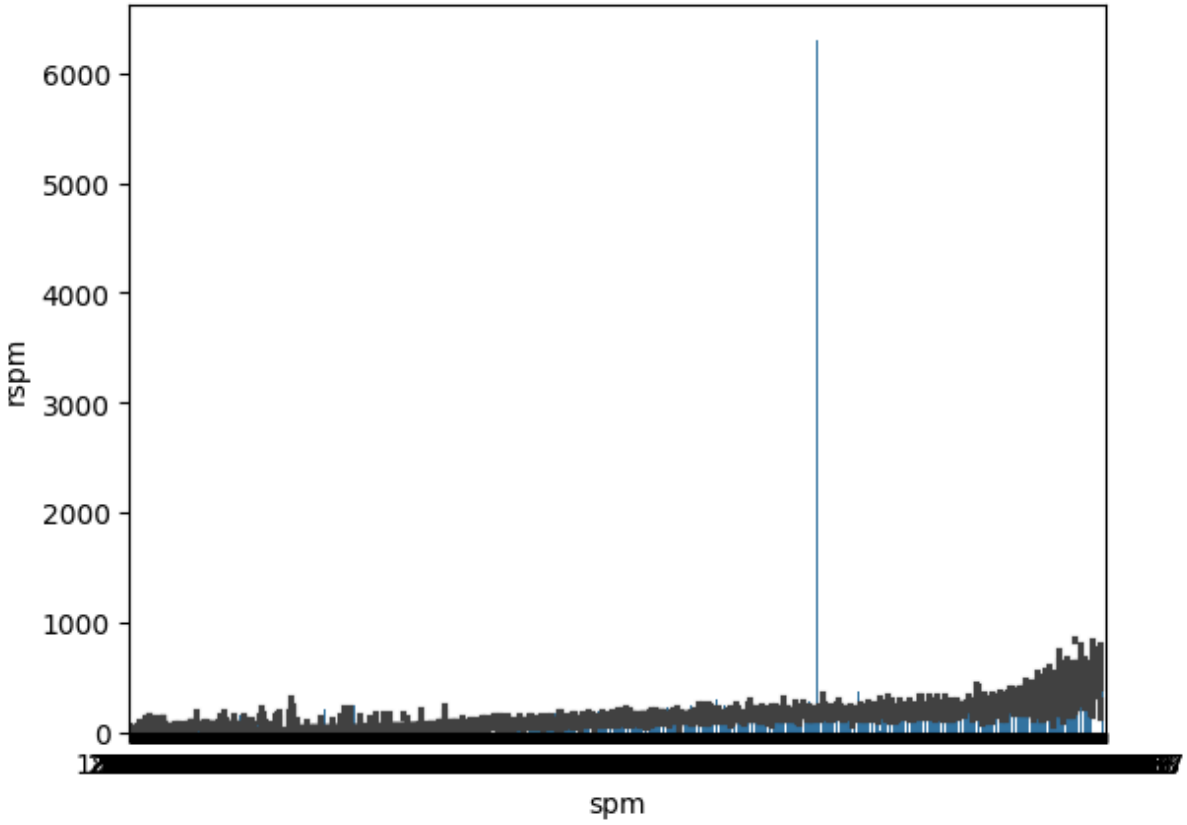
In [35]:
```
df.dtypes
```

Out[35]:
```
stn_code                       float64
sampling_date                  float64
state                          float64
location                       float64
agency                         float64
type                           float64
so2                            float64
no2                            float64
rspm                           float64
spm                            float64
location_monitoring_station    float64
pm2_5                          float64
date                           float64
dtype: object
```

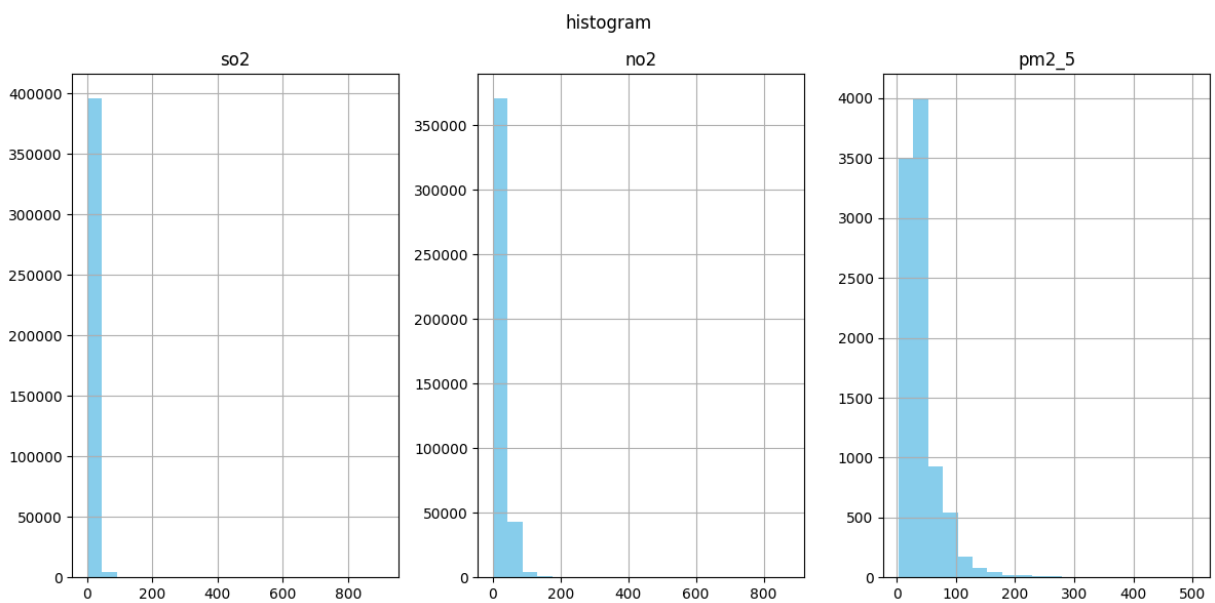
In [37]:
```
plt.figure(figsize=(8,6))
cols = ['so2', 'no2', 'rspm', 'spm']
corr = df[cols].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correaltion heatmap")
plt.show()
```



In [6]:
```
sns.barplot(x='spm', y='rspm', data=df)
plt.show()
```

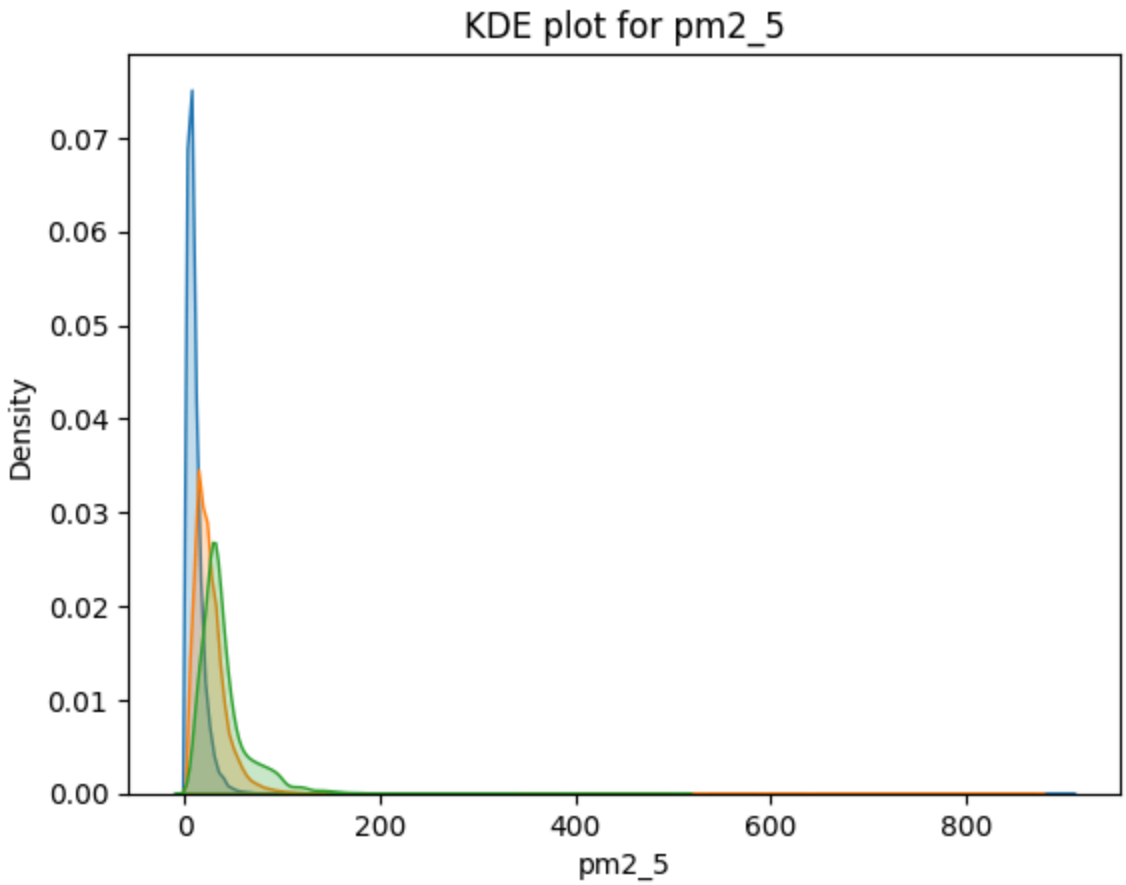


In [8]:
```
#Histogram
df[["so2","no2","pm2_5"]].hist(bins=20, figsize=(12,6), layout=(1,3), color='skyblu
plt.suptitle("histogram")
plt.tight_layout()
plt.show()
```

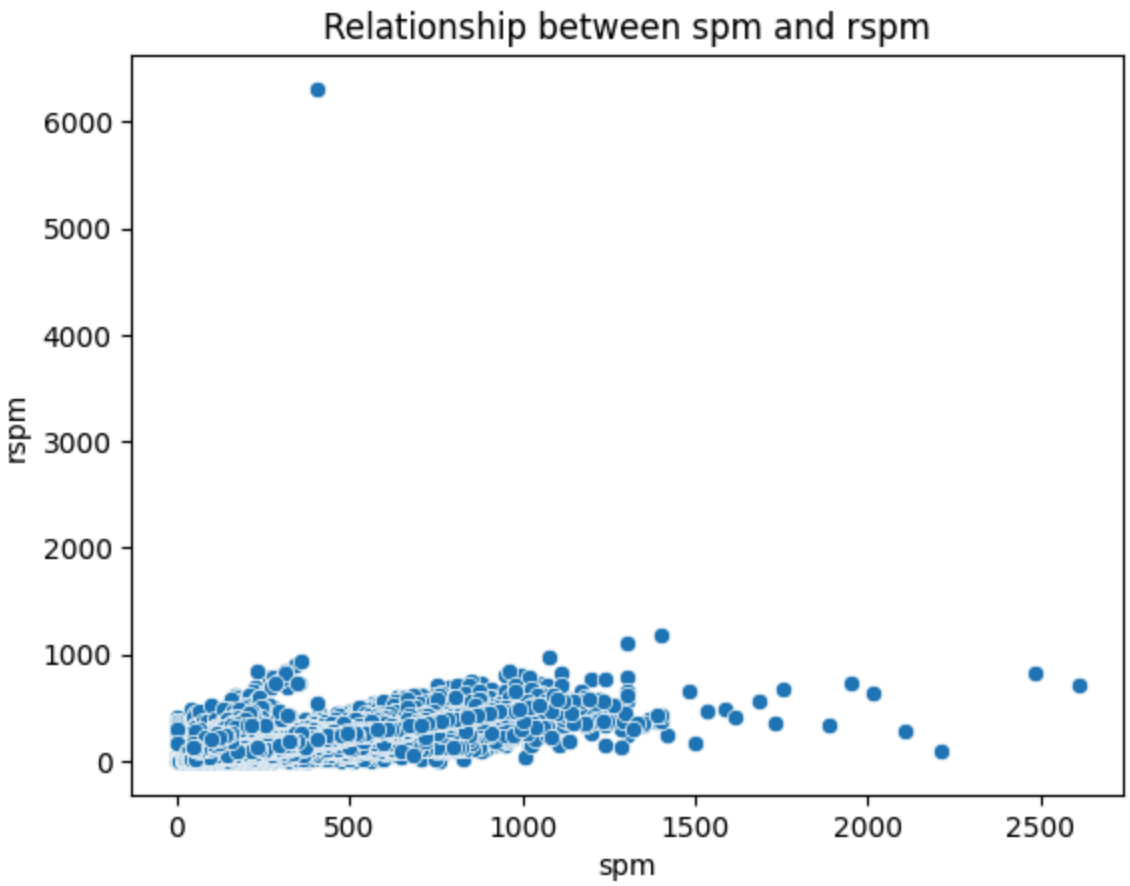


In [10]:
```
for column in ["so2","no2","pm2_5"]:
    sns.kdeplot(data=df[column], fill=True)
    plt.title(f"KDE plot for {column}")
    plt.xlabel(column)
```

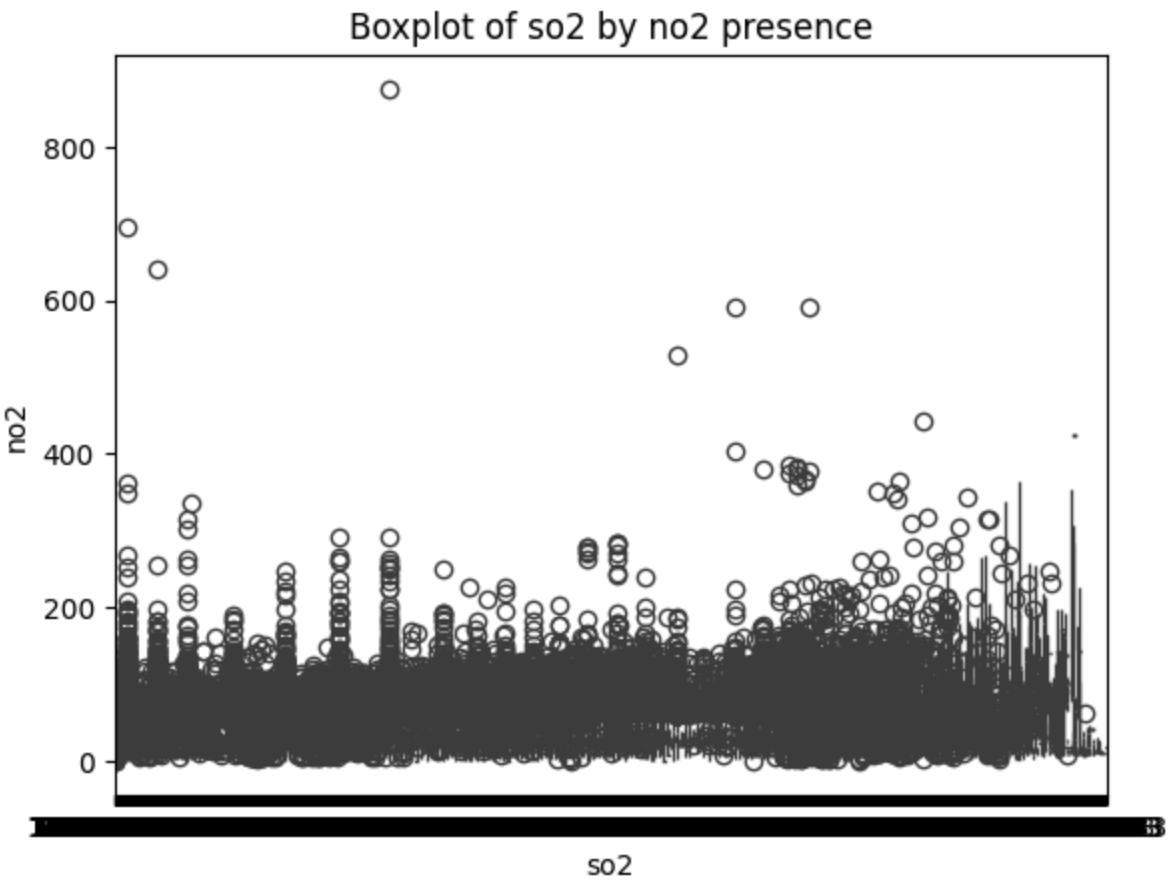
```
    plt.ylabel("Density")
plt.show()
```



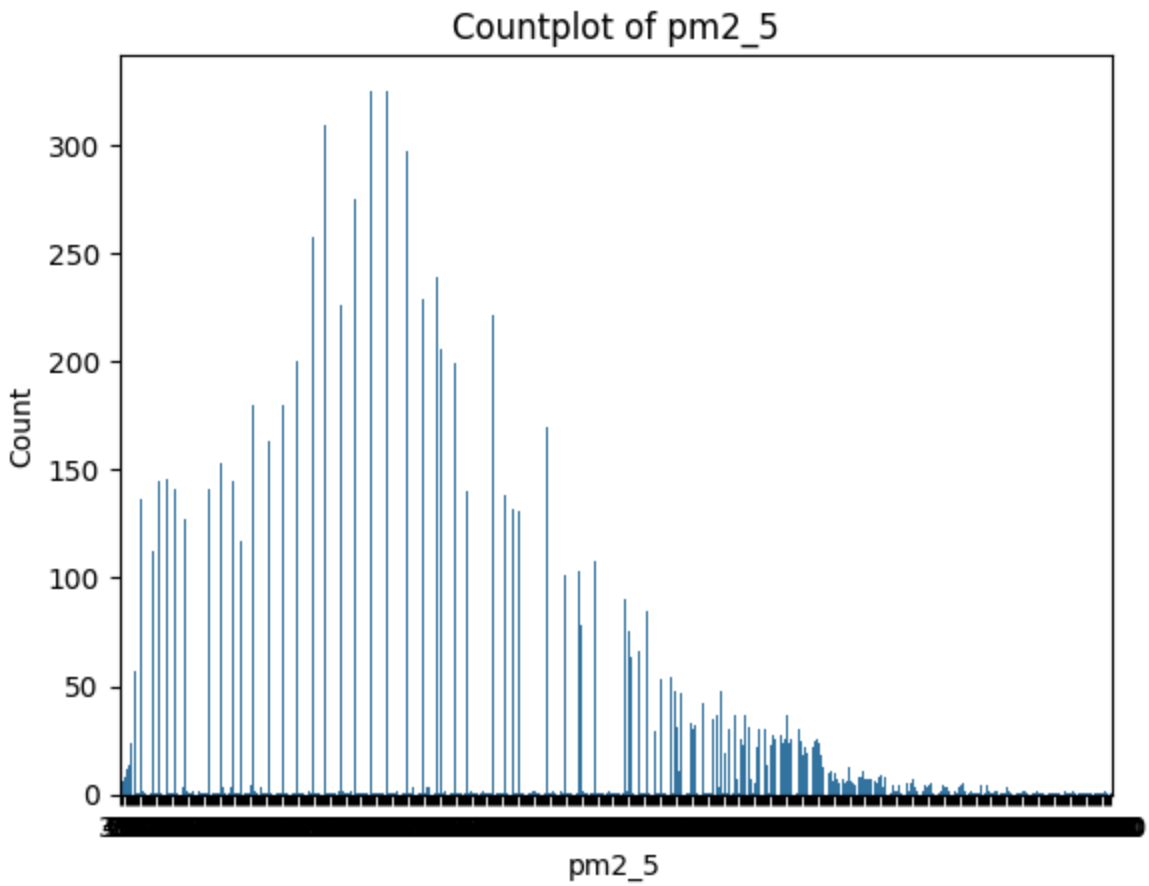
In [12]:
```
sns.scatterplot(x='spm', y='rspm', data=df)
plt.xlabel("spm")
plt.ylabel("rspm")
plt.title("Relationship between spm and rspm")
plt.show()
```



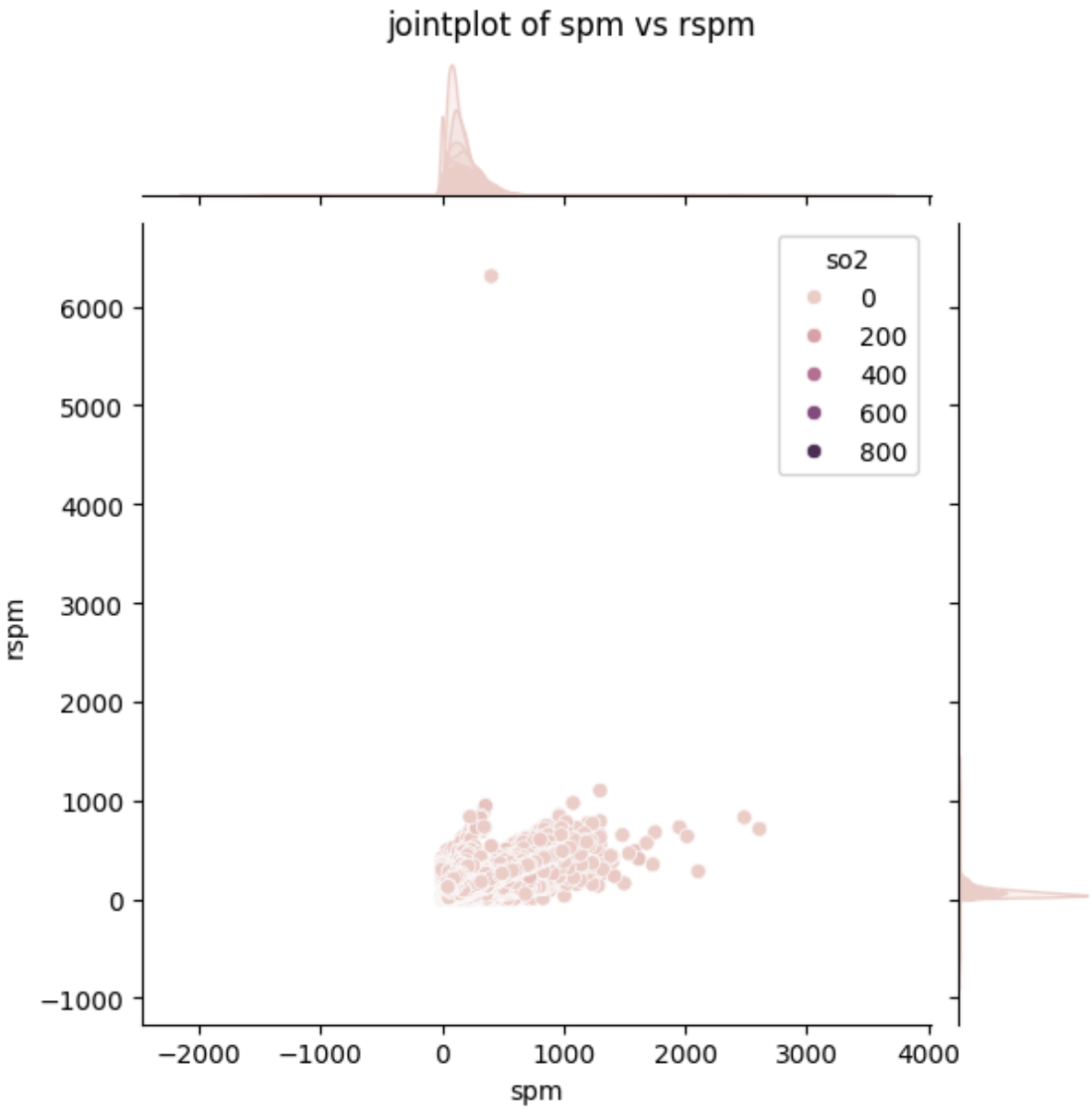
In [13]:
```
sns.boxplot(x='so2', y='no2', data=df)
plt.title("Boxplot of so2 by presence")
plt.show()
```



In [14]:
```
sns.countplot(x='pm2_5', data=df)
plt.title("Countplot of pm2_5")
plt.xlabel("pm2_5")
plt.ylabel("Count")
plt.show()
```



In [15]:
```
sns.jointplot(data=df, x='spm', y='rspm', kind='scatter', hue='so2')
plt.suptitle('jointplot of spm vs rspm',y=1.02)
plt.show()
```


jointplot of spm vs rspm
so2
0
200
400
600
800
rspm
spm
6000
5000
4000
3000
2000
1000
0
−1000
−2000
−1000
0
1000
2000
3000
4000